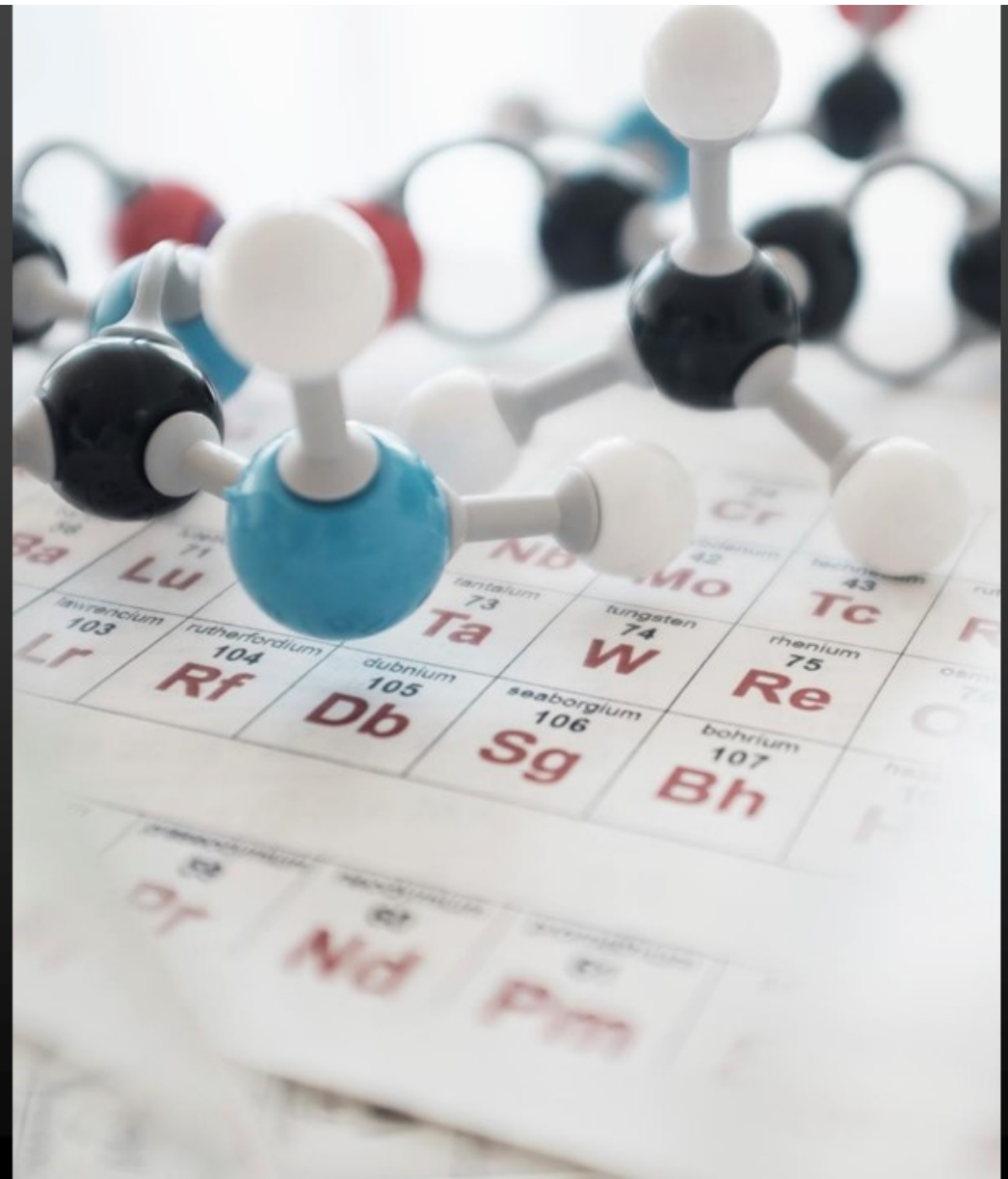




Opioids

Dr. Safaa Kishk



What are opioid analgesics?

Analgesics, or pain killers, that bind to opioid receptors which are found principally in the CNS and gastrointestinal tract.

There are a number of broad classes of opioids:

- **Natural opiates**

Alkaloids contained in the resin of the opium poppy including morphine, codeine and thebaine.

- **Semi-synthetic Opiates**

Created from the natural opioids such as hydromorphone, oxycodone and diacetylmorphine (heroin).

- **Fully synthetic opioids**

Fentanyl, methadone and tramadol.

- **Endogenous opioid peptides**

Produced naturally in the body, such as endorphins, enkephalins, dynorphins and endomorphins





Opium Poppy-*Papaver somniferum*



In general, opioids act on *mu-*, *delta-*, and *kappa-*receptors on CNS neurons producing analgesia via decreased neuronal transmitter release and decreased nociceptive impulse propagation.

Opioids appear to work by elevating the pain threshold, thus decreasing the brain's awareness of pain.

Receptor type	Location	Effects
μ	Brain, spinal cord	Analgesia, respiratory depression, euphoria, addiction, ALL pain messages blocked
κ	Brain, spinal cord	Analgesia, sedation, all non-thermal pain messages blocked
δ	Brain	Analgesia, antidepressant, dependence



From the previous tablet, we conclude that the three major types of opioid receptors differ in:

- 1. Cellular distribution.**
- 2. Relative affinity for various opioid ligands.**
- 3. Their contribution to specific opioid effects.**

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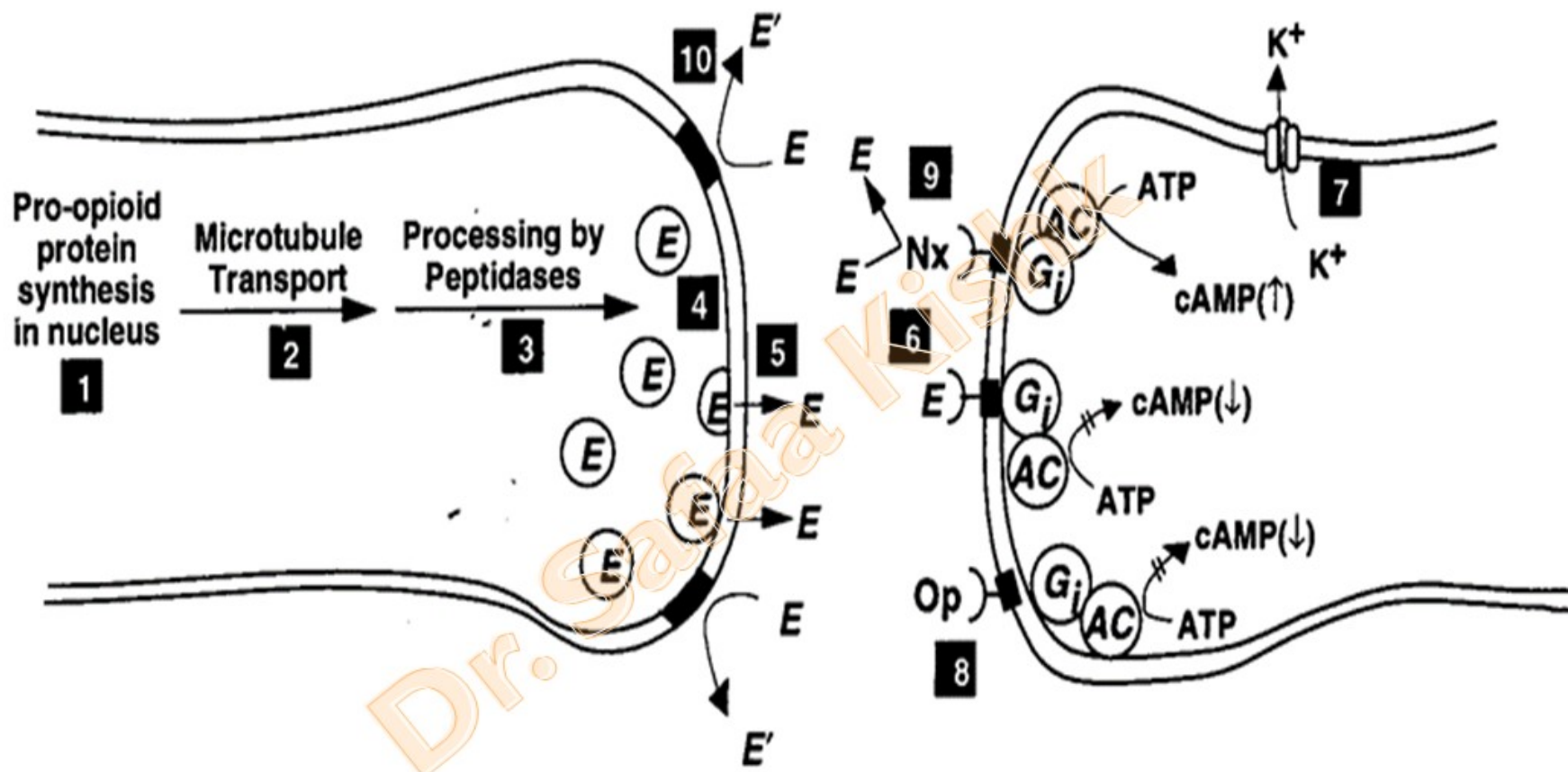
Enkephalin Properties:

- **Similar activity to the opioids (analgesia)**
- **Similar addiction and withdrawal effects.**
- **Enkephalins are antagonized by opioid antagonists (same receptor).**
- **Enkephalins are rapidly inactivated by specific peptidases in the brain.**

Details of Enkephalin Mechanism

- **Enkephalinergic system exists to modulate pain.**
- **Enkephalin release inhibits adenylate cyclase, decreasing cAMP levels and causing a K^+ efflux that hyperpolarizes the “pain neuron”, which inhibits nerve cell activity. Opioids also bind the enkephalin post-synaptic receptors.**





Dr. S. S. Vishal



μ is the analgesic receptor. μ_2 subtype is associated with respiratory depression and with GI receptors.

δ may be the antitussive receptor for codeine and related compounds. The antitussive actions of μ and δ specific agonists are antagonized by naloxone. However dextromethorphan sites don't bind codeine, and binding at these sites (likely σ sites) are not antagonized by naloxone. Therefore, there are at least two antitussive receptors.

κ is also analgesic. Binding site of several mixed agonist/antagonist compounds.

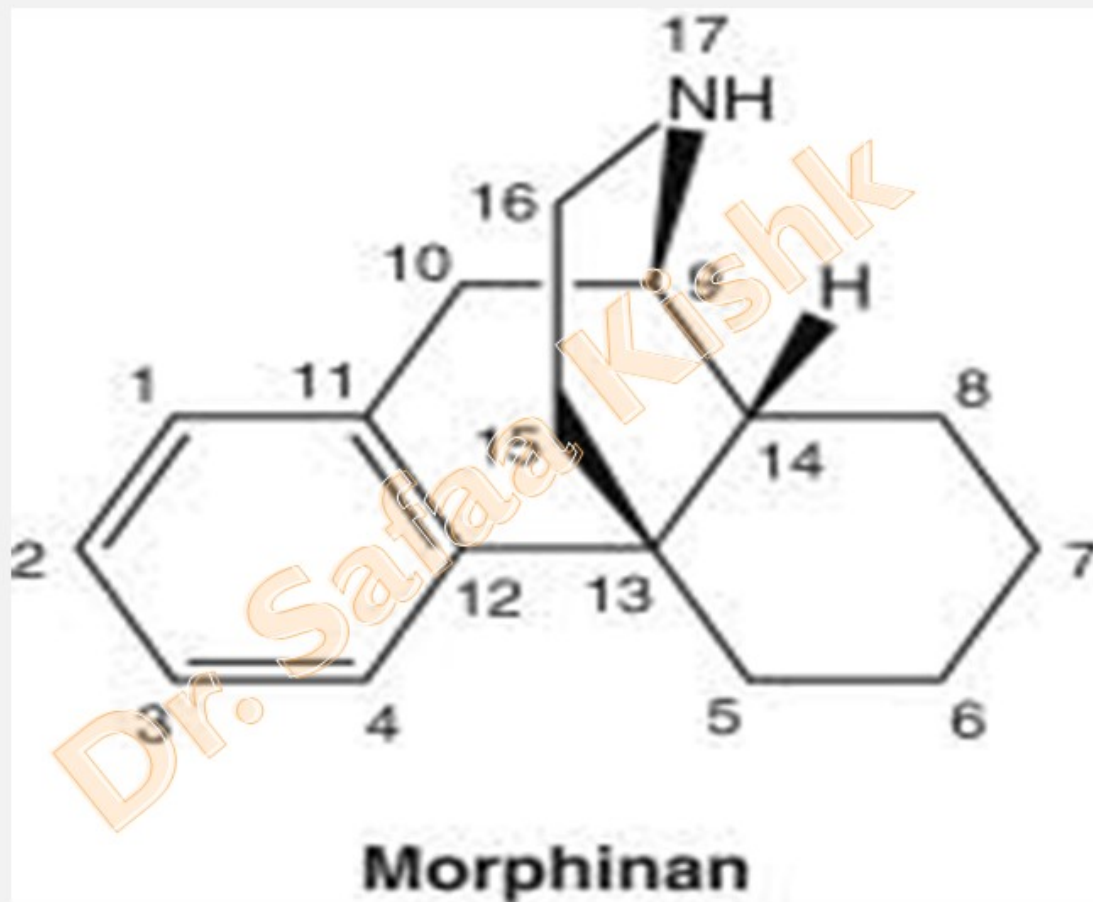
Pentazocine – agonist at κ , antagonist at μ

Buprenorphine – partial agonist at μ (slowly dissociates), antagonist at κ

Butorphanol – agonist at κ , antagonist at μ

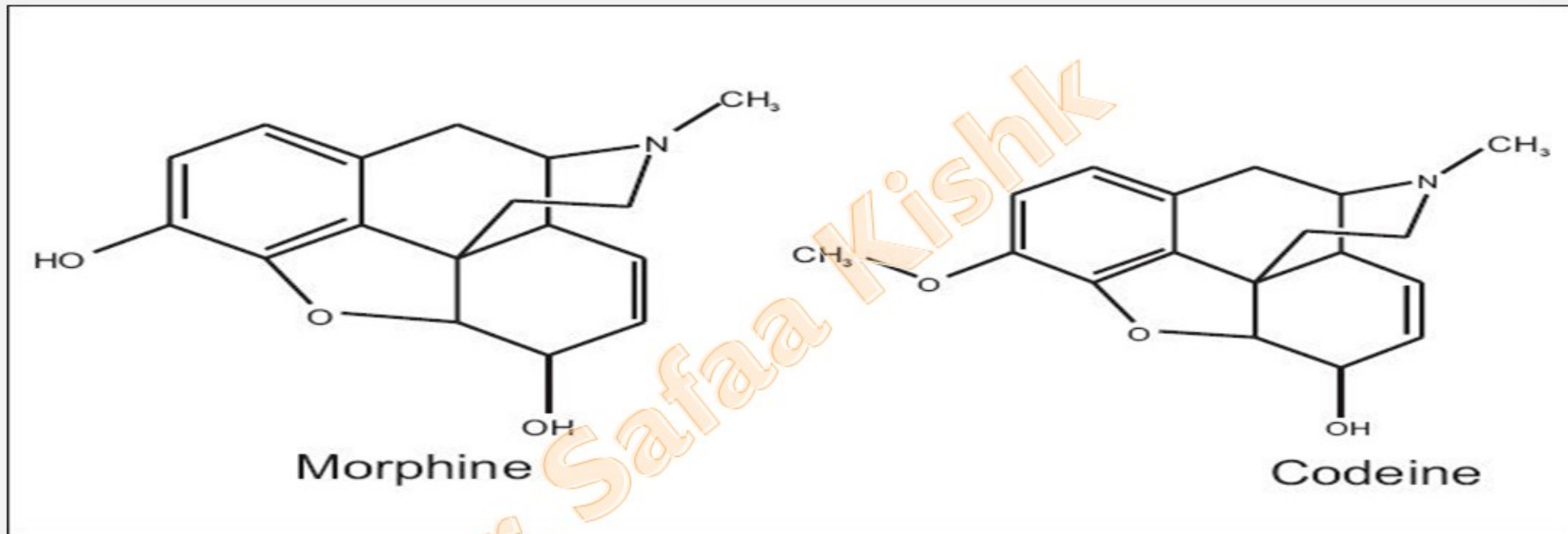


Morphinan ring



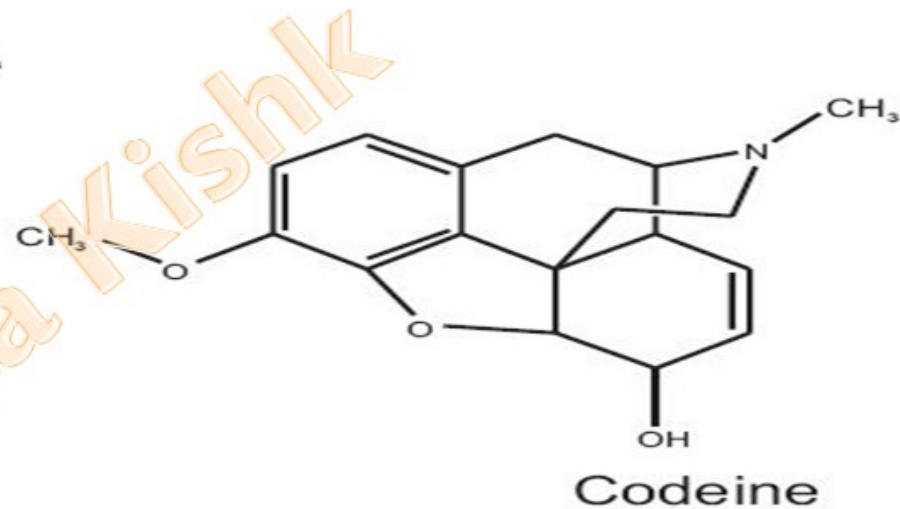
Natural opiates

Morphine



(5 α ,6 α)-17-methyl-7,8-didehydro-4,5-epoxymorphinan-3,6-diol

Codeine



(5 α ,6 α)-7,8-didehydro-4,5-epoxy- 3-methoxy-17-methylmorphinan-6-ol

Properties of Morphine

Good for treating dull, constant pain rather than sharp, periodic pain.

Unfortunately, it also has a large number of side effects including:

Depression of the respiratory center

Constipation

Excitation

Euphoria

Nausea

Pupil constriction

Tolerance and dependence

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Addiction

Clinical observation and experimental studies have indicated that a single exposure to morphine could induce tolerance and dependence.

Recent study shows that prolonged ventral tegmental area (VTA) dopamine neuron activities (DA) and opiate receptor desensitization followed single morphine exposure.

This may underlie the development of dependence and tolerance that may associate with the acute analgesic tolerance and acute addiction of morphine.



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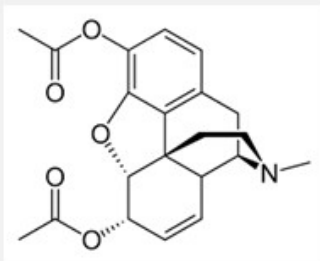
Opiophobia

The fear of prescribing opioid pain medications is known as "opiophobia".



Semisynthetic opiates

Heroin



(5 α ,6 α)-7,8-didehydro-4,5-epoxy-17-methylmorphinan-3,6-diol diacetate

Opioids of Abuse – Heroin, 2X more potent than morphine.

Diacetylmorphine is used as a recreational drug for the profound relaxation and intense euphoria it produces.

Users report an intense "rush" that occurs while the diacetylmorphine is being metabolized into 6-monoacetylmorphine (6-MAM) and morphine in the brain.

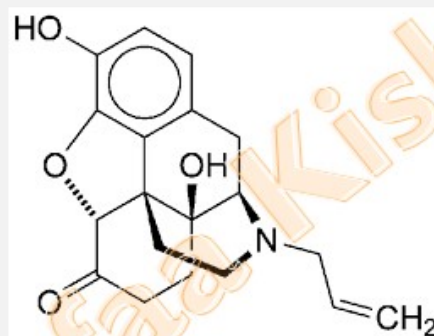
Heroin Overdose

A heroin overdose is usually treated with an opioid antagonist, such as **naloxone** (Narcan) which has high affinity for opioid receptors but does not activate them.

Many fatalities reported as overdoses are probably caused by interactions with other depressant drugs like alcohol or benzodiazepines.

Naloxone

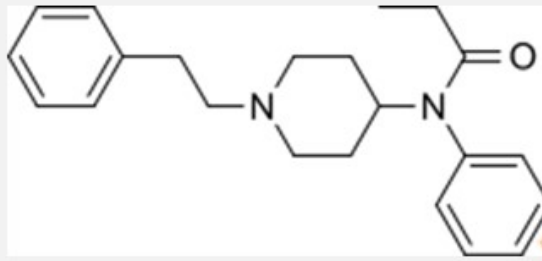
17-allyl-4,5 α -epoxy-3,14-dihydroxymorphinan-6-one



Naloxone has a high affinity for the μ -receptor, where it acts as an inverse agonist causing rapid removal of other drugs bound to these receptors.

Synthetic opiates

1. Fentanyl



1-Phenethyl-4-(N-phenylpropionamido)piperidine

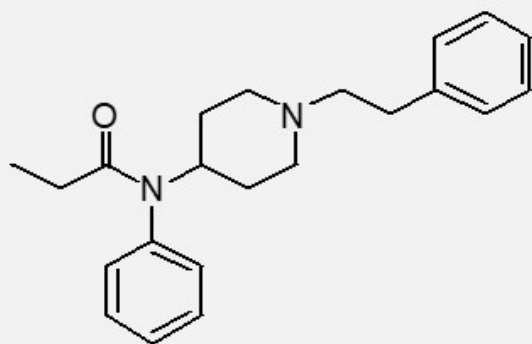
Fentanyl analogues

Alfentanil (Alfenta), an ultra-short acting (5-10 minutes) analgesic.

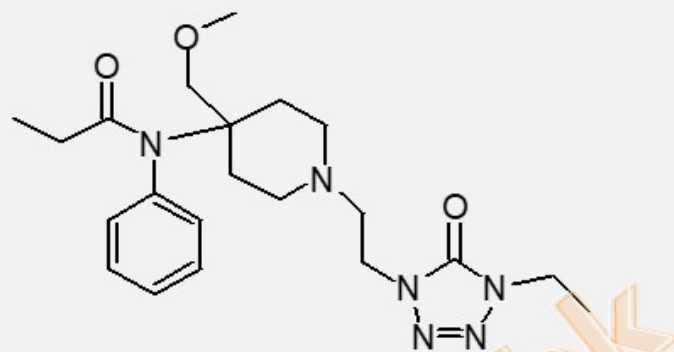
Sufentanil (trade name Sufenta), a potent analgesic (5 to 10 times more potent than fentanyl) for use in heart surgery.

Remifentanyl (trade name Ultiva), currently the shortest acting opioid, has the benefit of rapid offset, even after prolonged infusions.

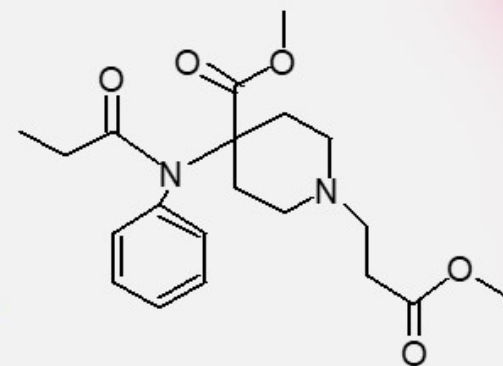
Carfentanyl (Wildnil) is an analogue of fentanyl with an analgesic potency 10,000 times that of morphine and used in veterinary practice to immobilize certain large animals such as elephants.



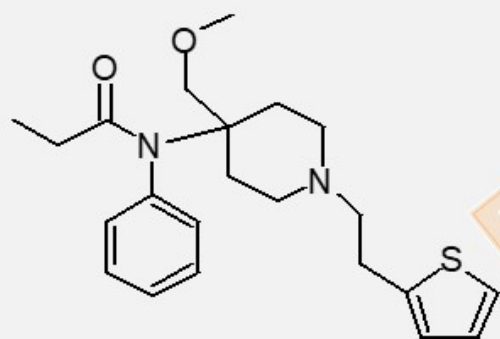
Fentanyl



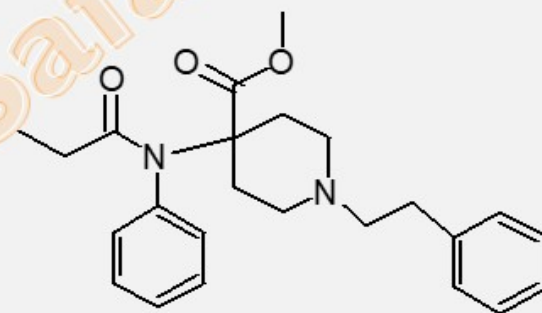
Alfentanil



Remifentanil

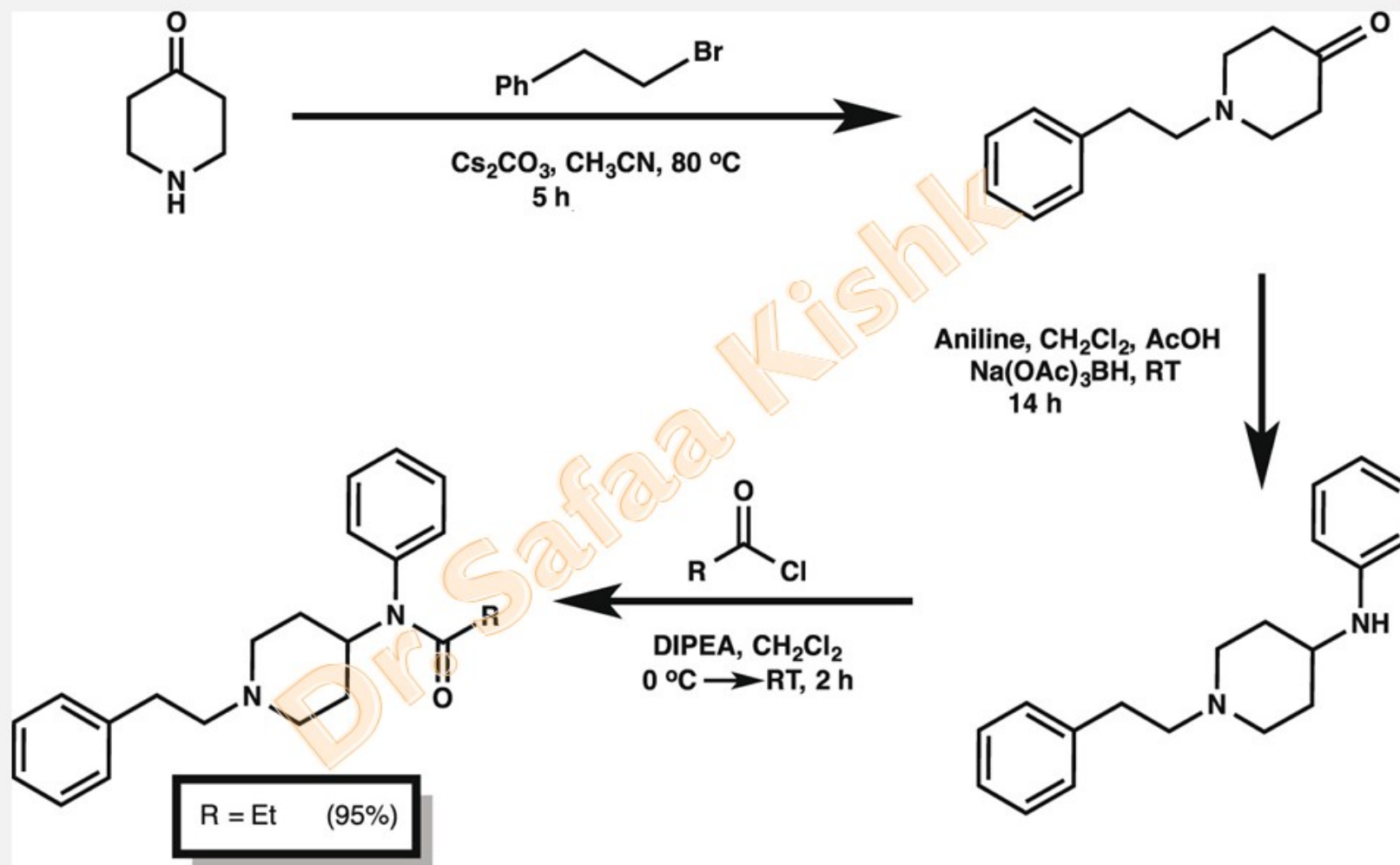


Sufentanil

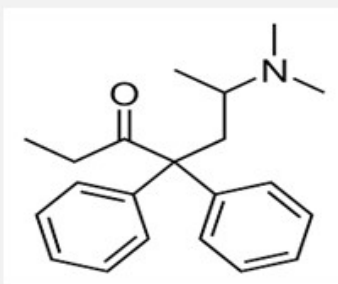


Carfentanil

Synthesis of fentanyl



2. Methadone

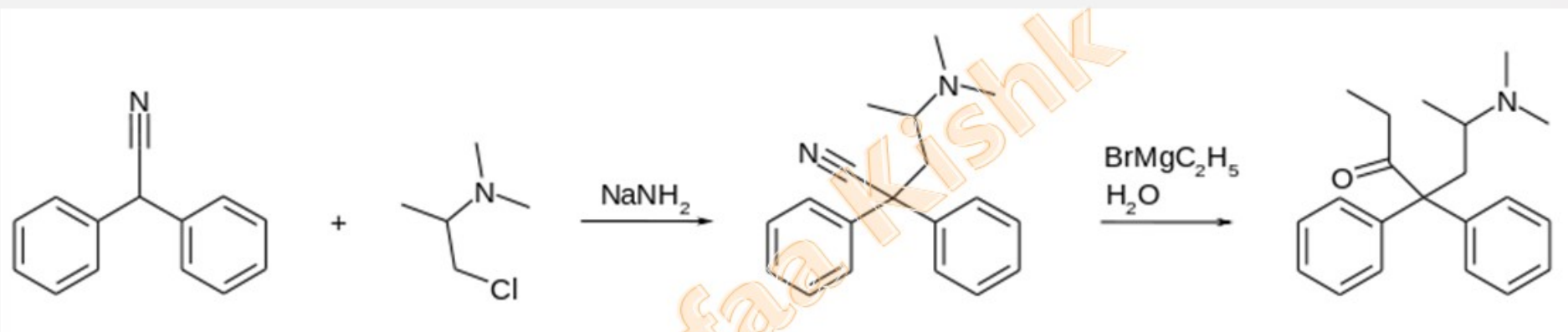


6-Dimethylamino-4,4-diphenyl-3-heptanone

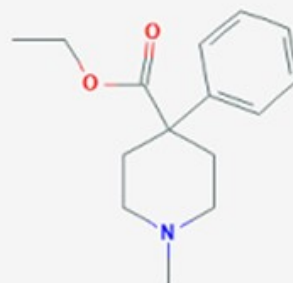
Methadone is a potent synthetic analgesic that works as a full μ -opioid receptor agonist.

Methadone shares similar effects and risks of other opioids such as morphine, hydromorphone, oxycodone, and fentanyl. However, it also has a unique pharmacokinetic profile. Compared with short-acting and even extended-release formulations of morphine, methadone displays a comparatively longer duration of action and half-life. These effects make methadone a good option for the treatment of severe pain and addiction as fewer doses are needed to maintain analgesia and prevent opioid withdrawal symptoms.

Synthesis of methadone



3. Meperidine (Pethidine)

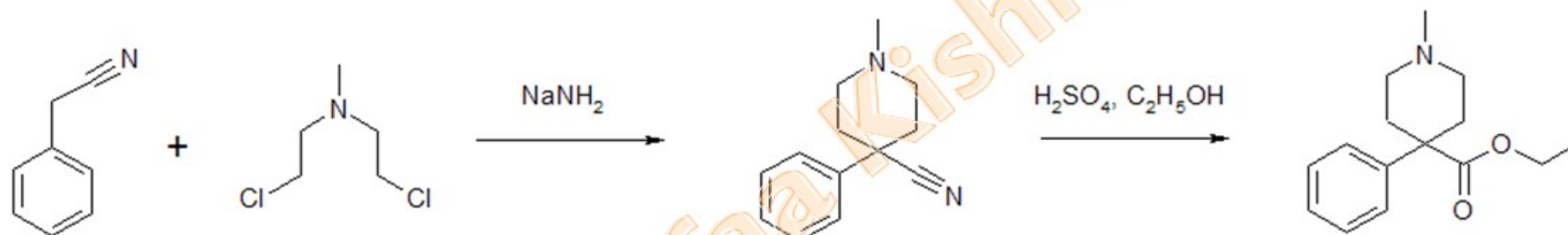


ethyl 1-methyl-4-phenylpiperidine-4-carboxylate

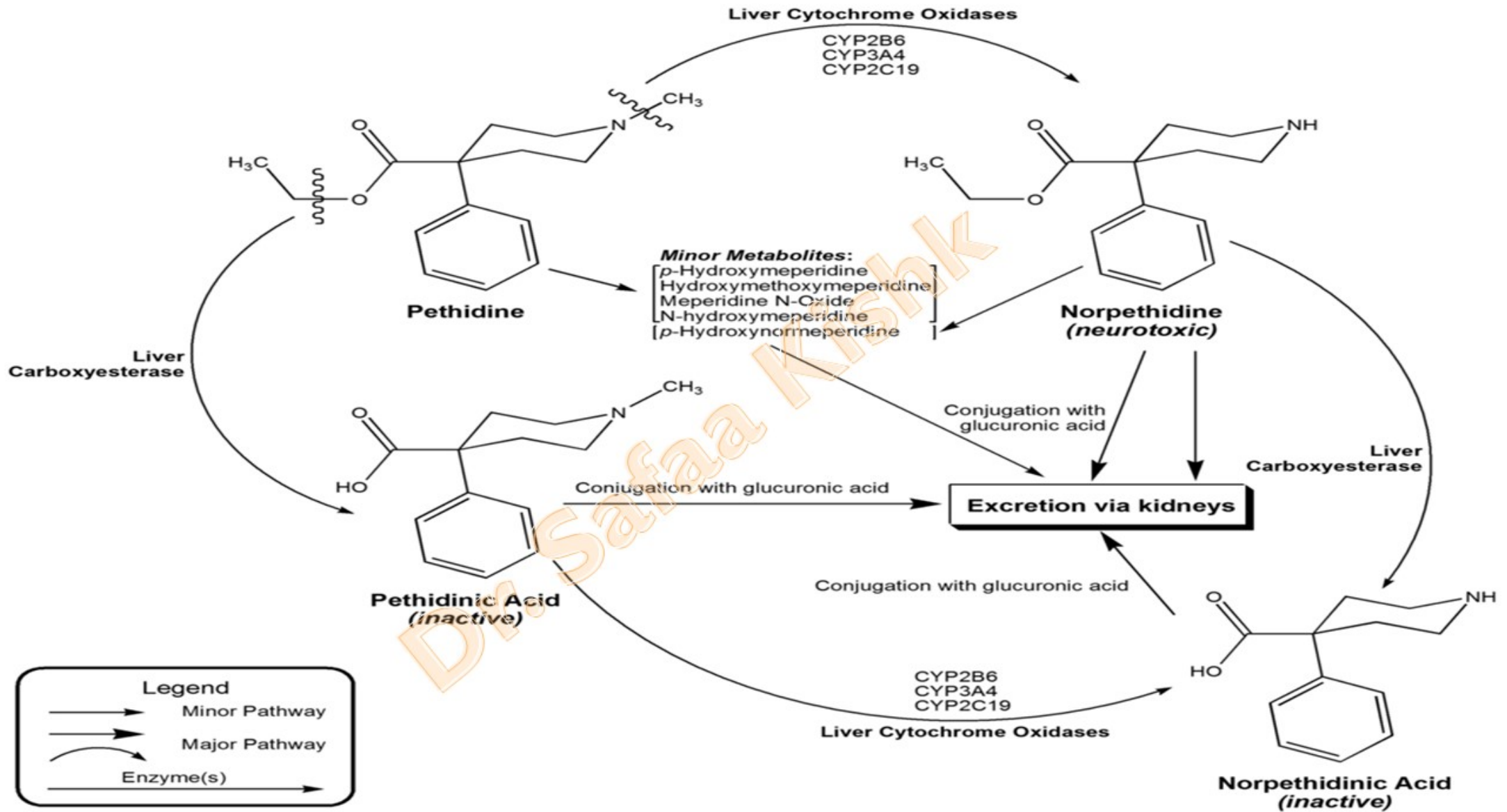
Meperidine is a synthetic piperidine ester with opioid analgesic activity. Meperidine mimics the actions of endogenous neuropeptides via opioid receptors, thereby producing the characteristic morphine-like effects on the mu-opioid receptor, including analgesia, euphoria, sedation, respiratory depression, bradycardia and physical dependence.



Synthesis of meperidine

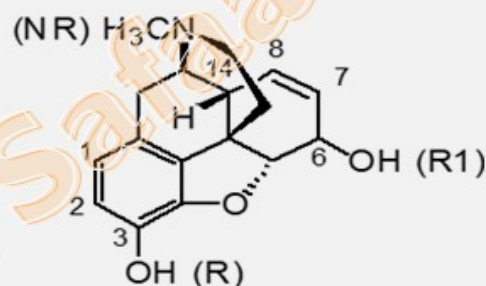


Metabolism of Pethidine



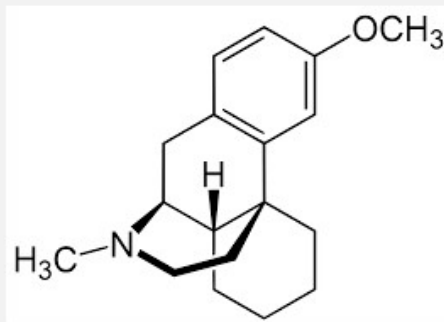
SAR:

1. Modifications at the phenolic OH (3-position) decrease analgesic potency, increase antitussive activity.
2. Modifications at the alcoholic OH (6-position) increase potency and addiction by increasing hydrophobicity.
3. Reduction of the 7,8-double bond: compounds with same substituents at R and R1 and a reduced double bond have the same potency as morphine. “dihydromorphine”, “dihydrocodeine.”



4. Modifications at Nitrogen. 2° N << 3°; 4° Nitrogen is inactive. -N(R) groups larger than CH₃ produce antagonists.

Dextromethorphan



The d-isomer of the codeine analog of levorphanol. Dextromethorphan shows high affinity binding to several regions of the brain, including the medullary cough center. This compound is an **NMDA receptor antagonist** (receptors, N-methyl-D-aspartate) and acts as a non-competitive channel blocker. It is one of the widely used antitussives, and is also used to study the involvement of glutamate receptors in neurotoxicity.

Medical uses

- Cough suppression

The primary use of dextromethorphan is as a cough suppressant, for the temporary relief of cough caused by minor throat and bronchial irritation (such as commonly accompanies the flu and common cold), as well as those resulting from inhaled particle irritants.

- Neuropsychiatric disorders

In 2013, a randomized clinical trial was conducted at Tabriz University of Medical Sciences in Tabriz, Iran, that indicated dextromethorphan may help reduce the overall discomfort and duration of withdrawal symptoms associated with opioid use disorder. When combined with clonidine, dextromethorphan reduced the overall time needed for withdrawal symptoms to peak by 24 hours while also reducing the severity of the symptoms compared to clonidine alone.



Thank You

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